



FPT UNIVERSITY

Carwash: Smart Car Wash Management System Software Requirement Specification

Project Code: Carwash

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Record of change

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21/07/2026	All sections	A	Create the Carwash SRS based on the provided FPT SRS template.	V1.0

SIGNATURE PAGE

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1 Introduction

The introduction of this Software Requirements Specification (SRS) provides an overview of the Carwash system. The document defines the main requirements for a web-based car wash booking, operations, payment, promotion, notification, and reporting platform.

This document follows the structure of the provided SRS template and is intentionally kept concise for project demonstration and review.

1.1 Purpose

The purpose of this SRS is to describe the required behavior of Carwash for customers, staff, managers, and administrators. It gives the development team a common baseline for implementation and testing.

1.2 Scope

Carwash allows customers to manage vehicles, book car wash services, pay online, follow booking status, receive notifications, and view history. Internal users use staff, manager, and admin workspaces to check in vehicles, manage wash sessions, assign staff, configure services and promotions, and view reports.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirement Specification
Customer	User who books and pays for car wash services
Staff	Employee who performs check-in and wash session tasks
Manager	User who supervises daily operations
Admin	System user who manages accounts, services, promotions, and reports
VNPay	Online payment gateway used by the project

1.4 References

- IEEE 830-1998 Recommended Practice for Software Requirements Specifications.
- Carwash project source code and implementation notes.
- Team UI and API requirements for customer, staff, manager, and admin modules.

1.5 Overview

Section 2 describes the overall product. Section 3 lists functional requirements and use cases. Section 4 defines non-functional requirements. Section 5 contains supporting information.

2 Overall Description

2.1 Product Perspective

Carwash is a full-stack web application with a browser-based frontend and a backend API. It centralizes the car wash service journey from customer booking to operational completion.

2.2 Product Functions

- Customer booking, vehicle management, payment, history, loyalty, and notifications.
- Staff check-in, session processing, status update, and task history.
- Manager operations queue, staff assignment, reports, notification handling, and settings.
- Admin account, service, booking, promotion, discount, package, review, and dashboard management.

2.3 User Characteristics

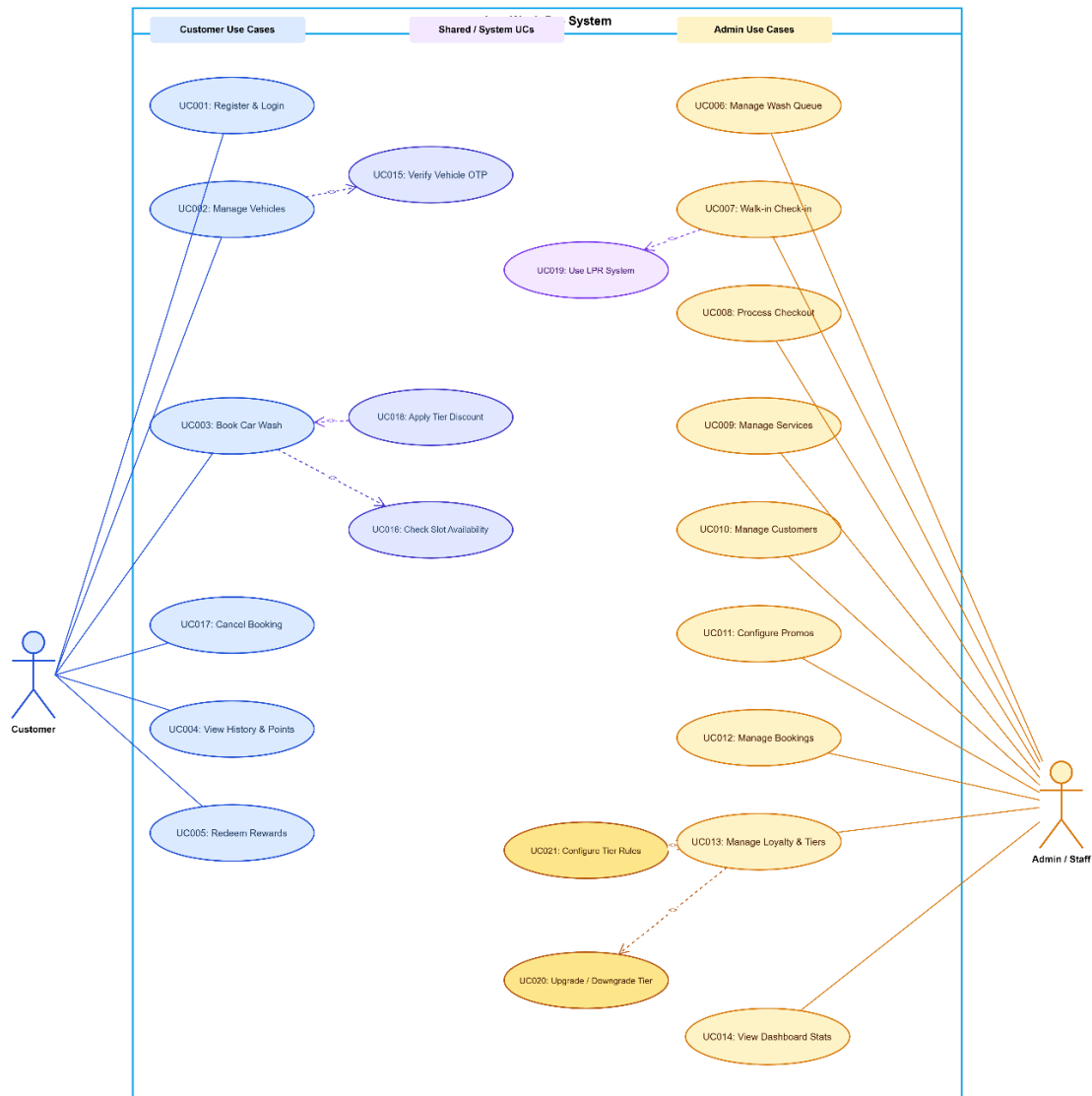
User Class	Main Activities
Customer	Book services, pay, track status, receive notifications, and review service history.
Staff	Check in vehicles, process assigned sessions, and update service progress.
Manager	Monitor queue, assign staff, resolve exceptions, and view operational reports.
Admin	Manage system data, accounts, business configuration, and analytics.

2.4 Constraints

- The system requires browser access and backend API availability.
- Role-based permissions must separate customer, staff, manager, and admin functions.
- Payment features depend on the configured VNPay environment.

3 FUNCTIONAL Requirements

3.1 <Use Cases Diagram>



Legend

- Association (Actor ↔ UC)
- .-> (mandatory sub-flow)
- .-> (optional conditional flow)
- Blue = Customer UCs
- Amber = Admin UCs
- Indigo = Sub-flow UCs

3.2 <Use Case Specifications>

Use-case No.	Use-case Name	Actor	Priority
UC001	Register & Login	Customer / Staff / Manager / Admin	High

Use-case No.	Use-case Name	Actor	Priority
UC002	Manage Vehicles	Customer	High
UC003	Book Car Wash	Customer	High
UC004	View History & Notifications	Customer	Normal
UC005	Process Payment	Customer	High
UC006	Manage Operations Queue	Manager	High
UC007	Check-in Vehicle	Staff	High
UC008	Update Wash Session	Staff	High
UC009	Manage Staff	Manager	High
UC010	Manage Services & Promotions	Admin / Manager	Normal

UC001 - Register & Login

USE CASE-001 SPECIFICATION				
Use-case No.	UC001	Use-case Version		1.0
Use-case Name	Register & Login			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	All users			
Brief Description	User signs up or logs in and is redirected to the correct workspace.			
Pre-condition	User has valid account information.			
Basic Flow	Enter credentials, verify if required, submit login, and open role-based dashboard.			
Post-condition	User session is created.			

UC002 - Manage Vehicles

USE CASE-002 SPECIFICATION				
Use-case No.	UC002	Use-case Version		1.0

Use-case Name	Manage Vehicles			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Customer			
Brief Description	Customer adds and maintains vehicle information used for booking.			
Pre-condition	Customer is logged in.			
Basic Flow	Open vehicle page, enter plate and vehicle details, then save.			
Post-condition	Vehicle data is stored.			

UC003 - Book Car Wash

USE CASE-003 SPECIFICATION				
Use-case No.	UC003	Use-case Version		1.0
Use-case Name	Book Car Wash			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Customer			
Brief Description	Customer creates a booking for an available service and schedule.			

Pre-condition	Customer has at least one vehicle.
Basic Flow	Select service, vehicle, time slot, review price, apply discount if available, and confirm.
Post-condition	Booking is created.

UC004 - View History & Notifications

USE CASE-004 SPECIFICATION				
Use-case No.	UC004	Use-case Version		1.0
Use-case Name	View History & Notifications			
Author	Carwash Development Team			
Date	21/07/2026	Priority		Normal
Actor	Customer			
Brief Description	Customer reviews booking history and system notifications.			
Pre-condition	Customer is logged in.			
Basic Flow	Open history or notification page, filter items, and view details.			
Post-condition	Information is displayed and read state may be updated.			

UC005 - Process Payment

USE CASE-005 SPECIFICATION				
Use-case No.	UC005	Use-case Version		1.0
Use-case Name	Process Payment			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Customer			
Brief Description	Customer pays for a booking using supported payment method.			
Pre-condition	Booking is valid and payment is required.			
Basic Flow	Choose payment method, submit payment, return from gateway, and view result.			
Post-condition	Payment status is updated.			

UC006 - Manage Operations Queue

USE CASE-006 SPECIFICATION				
Use-case No.	UC006	Use-case Version		1.0
Use-case Name	Manage Operations Queue			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Manager			

Brief Description	Manager monitors sessions and handles coordination alerts.
Pre-condition	Manager is logged in.
Basic Flow	Open operations queue, filter sessions, review details, prioritize or assign staff.
Post-condition	Queue and assignment data are updated.

UC007 - Check-in Vehicle

USE CASE-007 SPECIFICATION				
Use-case No.	UC007	Use-case Version		1.0
Use-case Name	Check-in Vehicle			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Staff			
Brief Description	Staff confirms arrival and starts the operation flow.			
Pre-condition	Booking exists and is not completed.			
Basic Flow	Search booking, verify vehicle/customer, and click Check-in.			
Post-condition	Booking moves to checked-in status.			

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UC008 - Update Wash Session

USE CASE-008 SPECIFICATION				
Use-case No.	UC008	Use-case Version		1.0
Use-case Name	Update Wash Session			
Author	Carwash Development Team			
Date	21/07/2026	Priority		High
Actor	Staff			
Brief Description	Staff updates a wash session through active stages.			
Pre-condition	Session is assigned or available.			
Basic Flow	Open task, start wash, update progress, add notes, and complete session.			
Post-condition	Session timeline is updated.			

UC009 - Manage Staff

USE CASE-009 SPECIFICATION				
Use-case No.	UC009	Use-case Version		1.0
Use-case Name	Manage Staff			
Author	Carwash Development Team			

Date	21/07/2026	Priority		High
Actor	Manager			
Brief Description	Manager views staff availability, workload, KPI, and assigns tasks.			
Pre-condition	Manager has staff permission.			
Basic Flow	Open staff page, filter staff, view detail, and assign suitable staff.			
Post-condition	Staff workload is updated.			

UC010 - Manage Services & Promotions

USE CASE-010 SPECIFICATION				
Use-case No.	UC010	Use-case Version		1.0
Use-case Name	Manage Services & Promotions			
Author	Carwash Development Team			
Date	21/07/2026	Priority		Normal
Actor	Admin / Manager			
Brief Description	Authorized users manage service catalog, discounts, vouchers, and promotions.			
Pre-condition	User has management permission.			
Basic Flow	Create or edit service/promotion data and save.			

Post-condition	Business configuration is updated.

4 NON-FUNCTIONAL Requirements

4.1 Usability

- The user interface must be clear, responsive, and consistent across public, customer, staff, manager, and admin pages.
- Important actions must show loading, success, error, and empty states where appropriate.

4.2 Reliability

- Booking and payment status must remain consistent after refresh or navigation.
- Completed bookings must not be reassigned or handed over to another active task.

4.3 Performance

- Main pages should load quickly under normal demo data volume.
- Filters and dashboards should update without freezing the interface.

4.4 Supportability

- The project should use shared API services, reusable UI components, and consistent response handling.

4.5 Design Constraints

- Frontend must communicate with backend APIs instead of using hard-coded production data.
- Sensitive configuration such as payment secrets must stay in backend environment variables.

5 Supporting Information

5.1 Main Data Objects

Object	Description
User	Account, profile, authentication, and role data.
Vehicle	Customer vehicle information used for booking.
Booking	Reservation, service, schedule, customer, payment, and status data.
Wash Session	Operational state after check-in.
Service	Car wash package or add-on sold by the system.
Promotion	Discount or voucher campaign.
Notification	Customer or manager message.
Review	Customer service rating and feedback.

5.2 Approval

This document is used as the baseline Software Requirement Specification for the Carwash project version 1.0.